

## Project Executable Files

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|----------------------|---------------------------------------------------------------------------------------------------------|
| <b>Date</b>          | 15 March 2026                                                                                           |
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| <b>Project Name</b>  | EcoTrack - AI-Powered Carbon Footprint Tracker                                                          |
| <b>Maximum Marks</b> | 3 Marks                                                                                                 |

### Project Executable Files

This section requires submission of all executable and deployable files for the EcoTrack project. All files are organized, named, and available in the linked repository. The evaluator can run or deploy the solution using the provided files and instructions below.

#### Step 1: Submission Checklist

| S.No | Item to Submit                                                  | Submitted (Yes / No / NA) |
|------|-----------------------------------------------------------------|---------------------------|
| 1    | Complete source code (all files and folders)                    | Yes                       |
| 2    | README / Setup Guide (instructions to run the project)          | Yes                       |
| 3    | requirements.txt / package.json / dependency file               | Yes                       |
| 4    | Database schema / seed files (if applicable)                    | Yes                       |
| 5    | Environment configuration file (.env.example or similar)        | Yes                       |
| 6    | Deployed application URL (if hosted)                            | NA                        |
| 7    | APK / Executable binary (if applicable for mobile/desktop apps) | NA                        |
| 8    | Dockerfile / Containerization config (if applicable)            | No                        |
| 9    | Test files and test results                                     | Yes                       |
| 10   | Demo video or walkthrough (if required)                         | Yes                       |

## Step 2: File / Folder Structure

Overview of the EcoTrack project's file and folder structure:

```
EcoTrack-root/
  frontend/      - React.js (Vite) source code
    src/components/ - Reusable UI (HabitForm, CarbonGauge, BadgeCard)
    src/pages/    - Dashboard, History & Charts, AI Suggestions pages
    src/services/ - Axios API service layer
    package.json - Frontend dependencies
  backend/       - Flask backend
    app.py       - Main Flask application entry point
    routes/      - API route blueprints (habits, auth, suggestions)
    models/      - Database models (Firestore / SQLite fallback)
    requirements.txt - Python dependencies
    .env.example - Environment variable template
    README.md    - Setup and run instructions
```

## Step 3: Deployment / Access Details

| Field                              | Details                                                                                            |
|------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Hosted / Deployed URL</b>       | <a href="https://github.com/Praveen0550/EcoTrack">https://github.com/Praveen0550/EcoTrack</a>      |
| <b>Login Credentials (Demo)</b>    | Create a free account via the Sign-Up screen, or use demo account:<br>demo@ecotrack.com / Demo@123 |
| <b>Platform / Hosting Provider</b> | Local development environment (Flask dev server + Vite dev server)                                 |
| <b>Repository Link</b>             | <a href="https://github.com/Praveen0550/EcoTrack">https://github.com/Praveen0550/EcoTrack</a>      |
| <b>Demo Video Link</b>             | To be added prior to final submission                                                              |

#### Step 4: Run Instructions

Step-by-step instructions for the evaluator to run the project locally:

**Pre-requisites:**

Node.js v18+, Python 3.10+, npm, pip, a Firebase project (Auth + Firestore), a Groq API key, and a Carbon Interface API key.

Step 1: Clone the repository: `git clone https://github.com/Praveen0550/EcoTrack.git`

Step 2: Backend setup: `cd backend`, then `pip install -r requirements.txt`

Step 3: Configure environment: copy `.env.example` to `.env` and add your Firebase, Groq, and Carbon Interface API credentials

Step 4: Start the backend: `python app.py` (serves on `http://localhost:5000`)

Step 5: Frontend setup: `cd frontend`, then `npm install`

Step 6: Start the frontend: `npm run dev` (serves on `http://localhost:5173`)

Step 7: Open the browser at `http://localhost:5173`, sign up, and begin logging daily habits

#### Step 5: Known Issues / Limitations

| S.No | Known Issue / Limitation                                                              | Workaround / Status                                            |
|------|---------------------------------------------------------------------------------------|----------------------------------------------------------------|
| 1    | AI suggestion response time can be slower under load due to external Groq API latency | Asynchronous suggestion generation implemented as a workaround |
| 2    | Application is not yet deployed to a public hosting platform                          | Deployment to Render/Vercel planned for a future release       |
| 3    | Minor code duplication across travel/food/energy form handlers                        | Refactoring into a shared reusable form component is planned   |